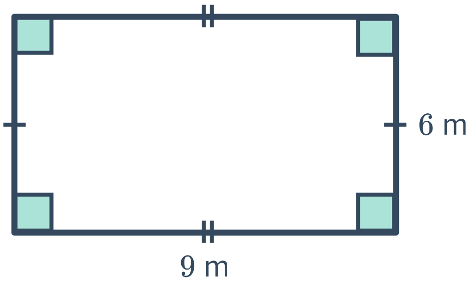
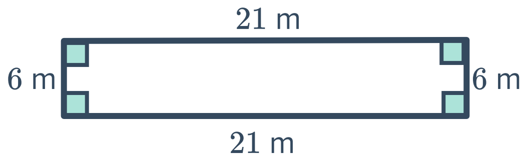
Perimeter of a rectangle

1 Find the perimeter of the following rectangles:

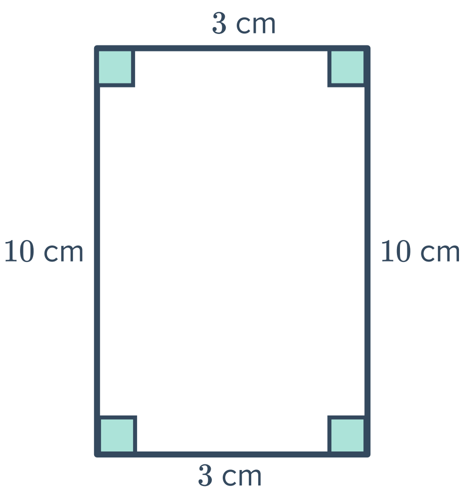
a



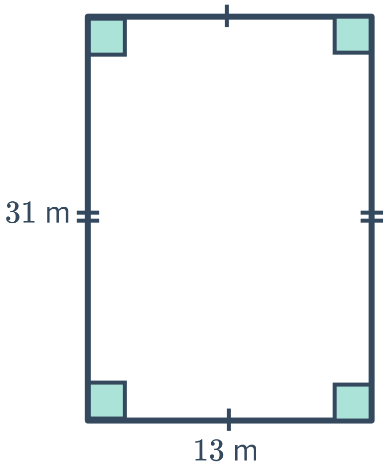
b



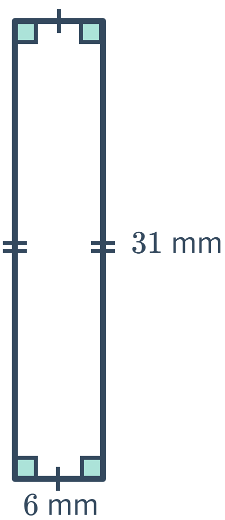
c



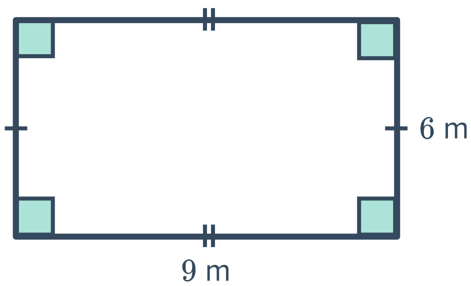
d



e



f



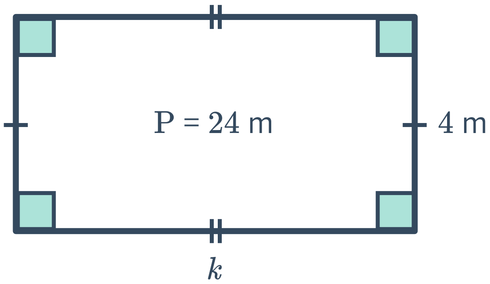
2 Find the perimeter of a rectangle given th[illegible]owing dimensions:

a Length: $20\ \mathrm{m}$, Width: $7\ \mathrm{m}$

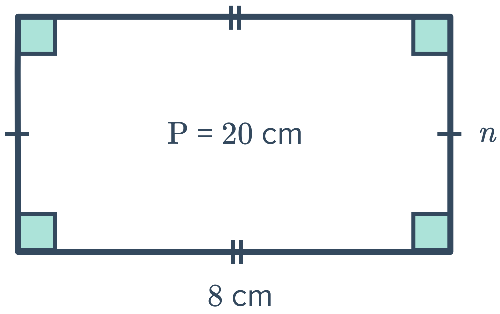
b Length: $2\ \mathrm{m}$, Width: $13\ \mathrm{m}$

3 Given the perimeter, P, of the following rectangles, find the value of the pronumeral:

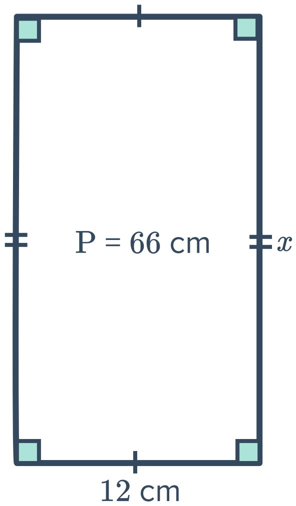
a



b



c

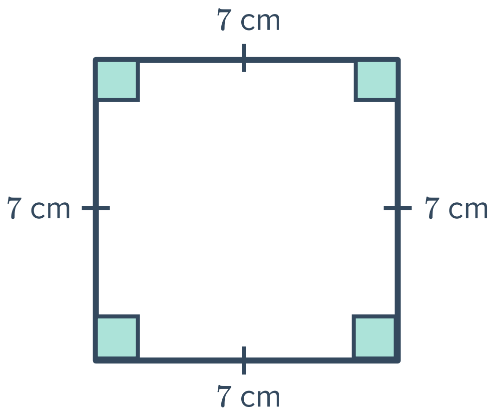


4 A rectangle has a perimeter of $26\ \mathrm{cm}$. If it has a length of $9\ \mathrm{cm}$, find its width.

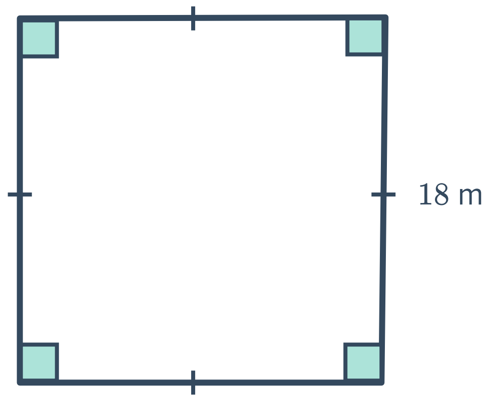
Perimeter of a square

5 Find the perimeter of the following squares:

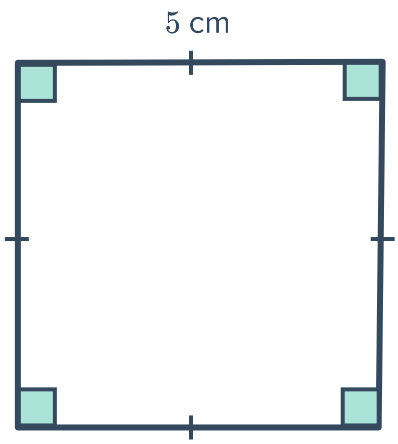
a



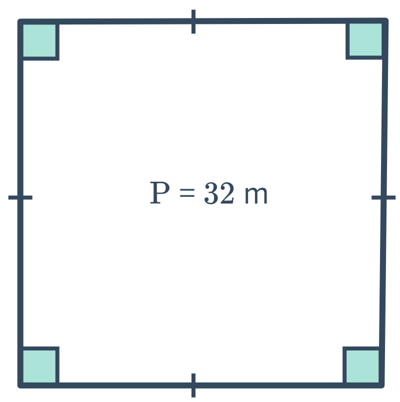
b



c



6 Find the side length of the square shown with perimerer of 32 m:

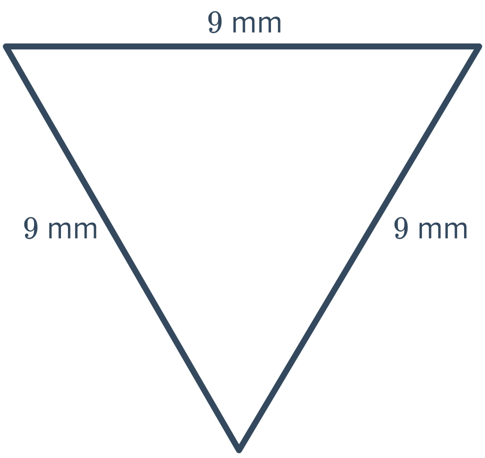


7 A square has a perimeter of 12 mm. Find the side length of the square.

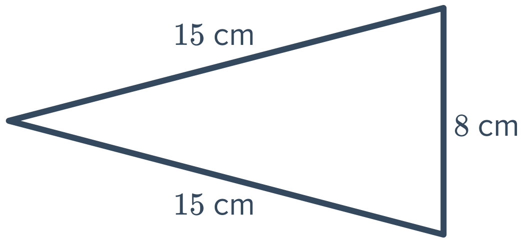
Perimeter of a triangle

8 Find the perimeter of the following triangles:

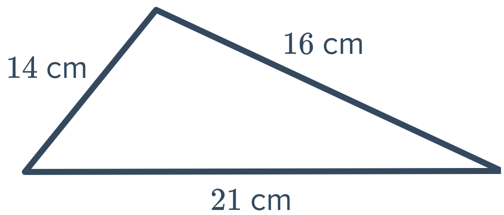
a



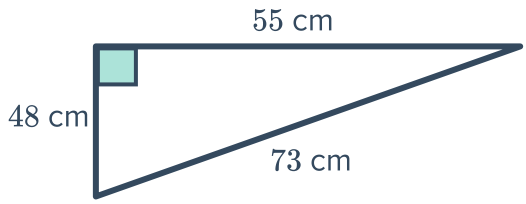
b



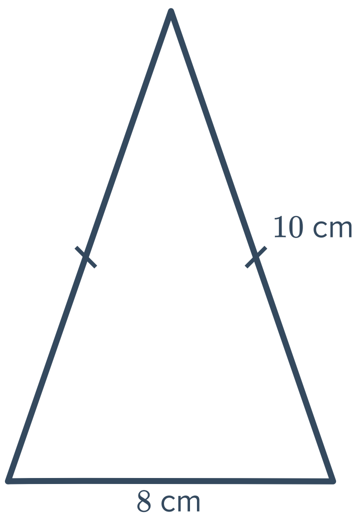
c



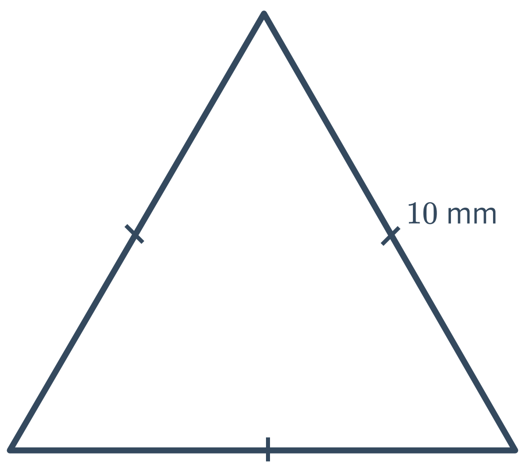
d



e



f



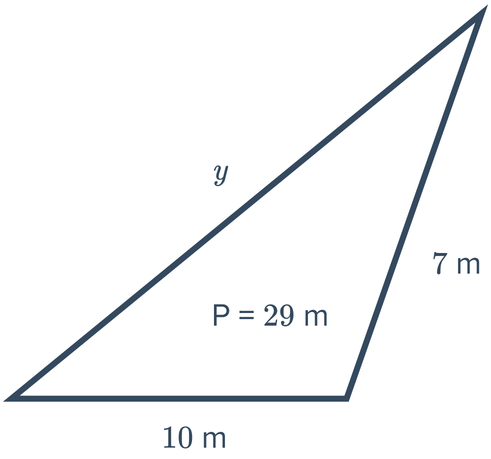
a A scalene triangle with side lengths 13 mm, 17 mm and 18 mm.

b An equilateral triangle with a side length of 4 m.

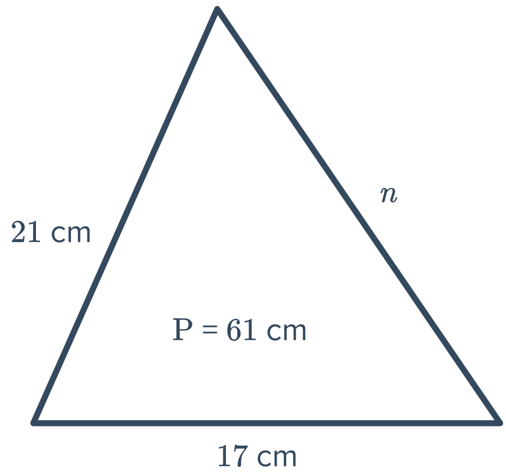
c An isosceles triangle where the 2 equal side lengths are 13 mm each and the third side measures 4 mm.

10 Given the perimeter, P, of the following rectangles, find the value of the pronumeral:

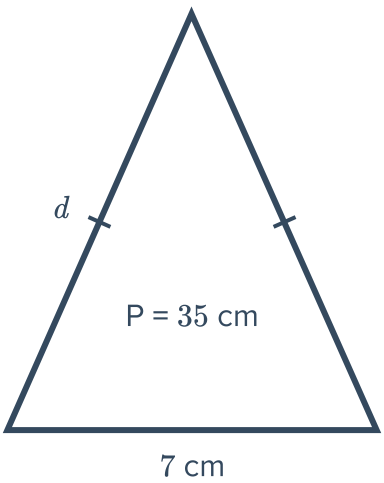
a



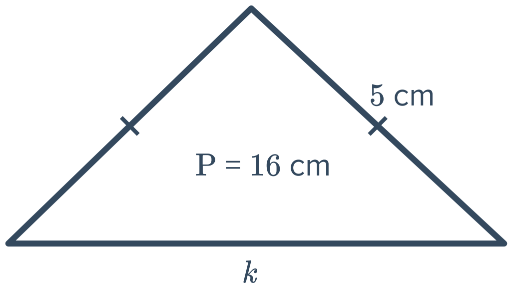
b



c



d

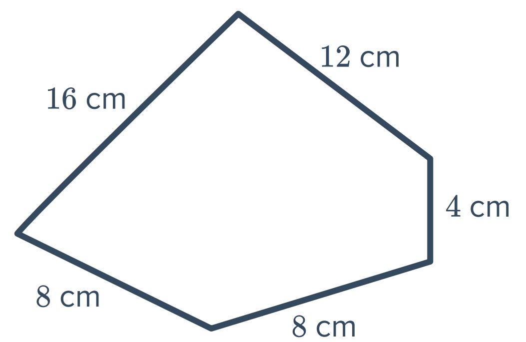


11 Find the side length of a triangle that has a perimeter of 18 m and two side lengths that both measure to be 5 m.

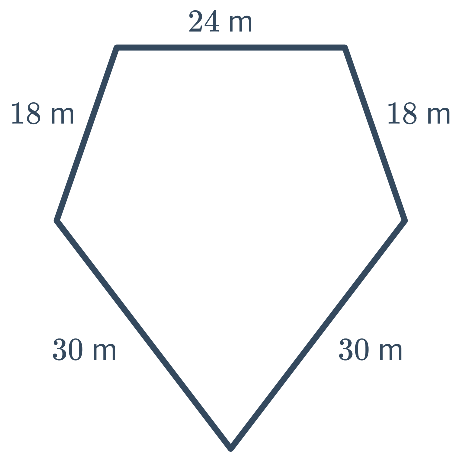
12 Find the side length of one of the equal sides of an isosceles triangle that has a perimeter of 22 cm, and a third side length that measures to be 4 cm.

13 Find the perimeter of the following composite shapes:

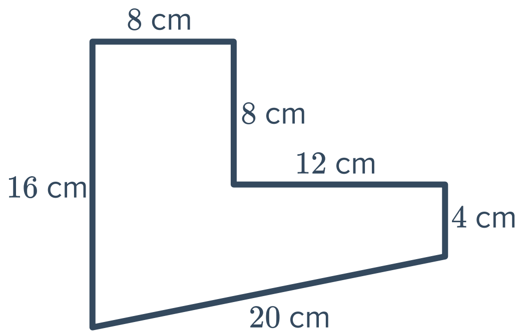
a



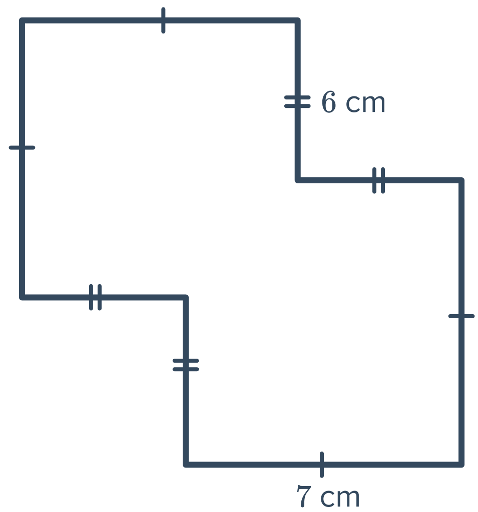
b



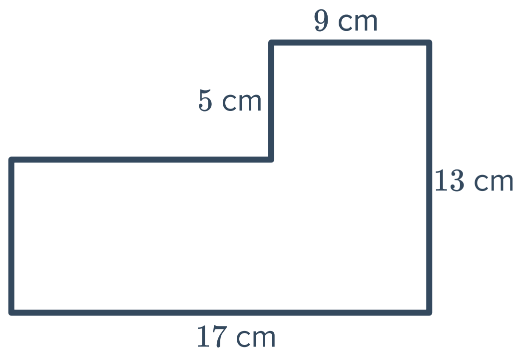
c



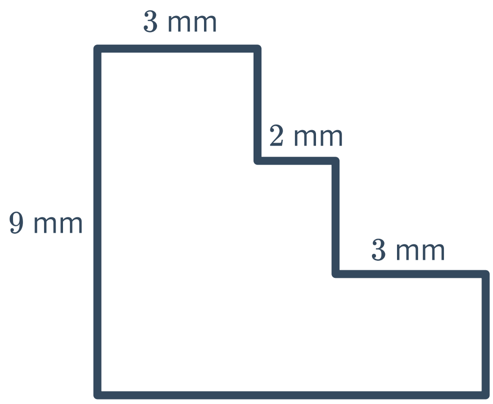
d



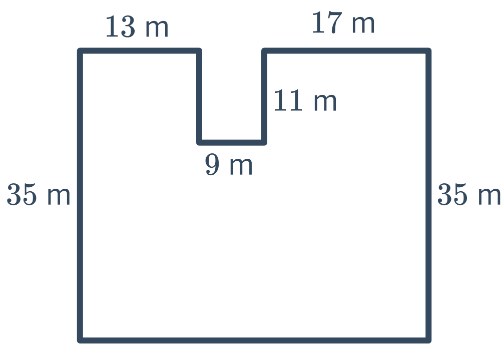
e



f



g



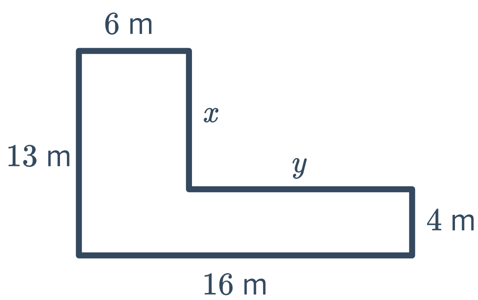
14 For each of the following figures:

i Find the length x.

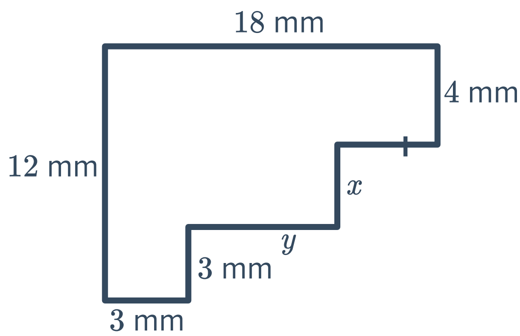
ii Find the length y.

iii Calculate the perimeter of the figure.

a



b



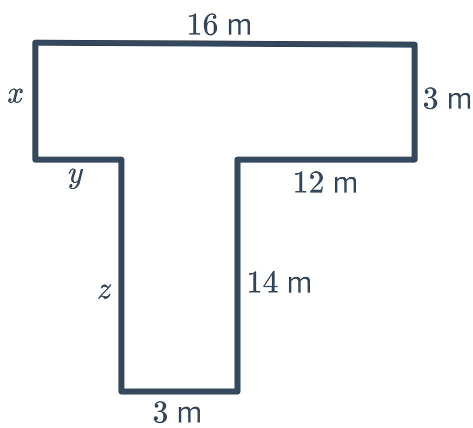
15 Consider the following figure:

a Find the length x.

b Find the length y.

c Find the length z.

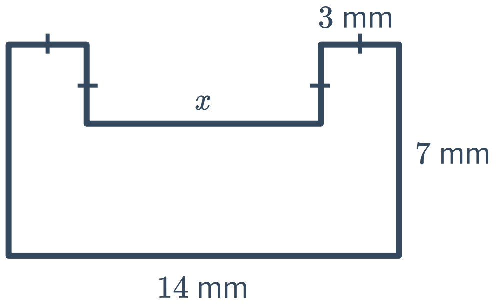
d Calculate the perimeter of the figure.



16 Consider the following figure:

a Find the length x.

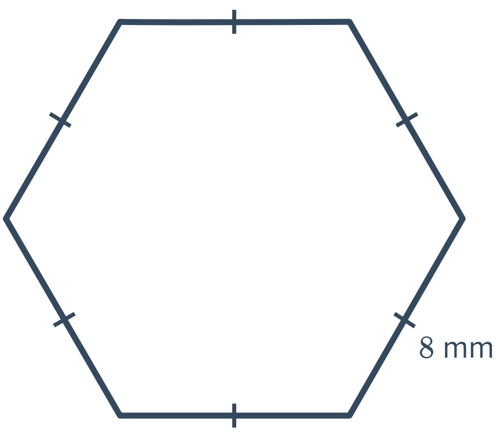
b Calculate the perimeter of the figure.



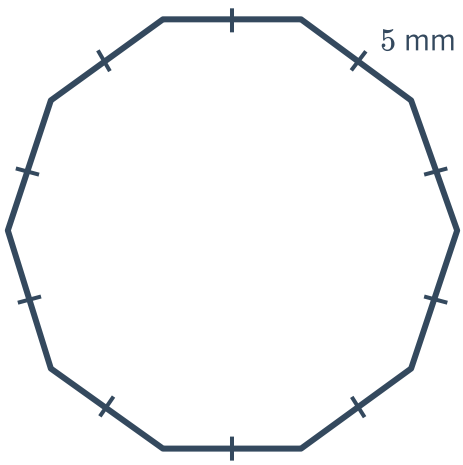
Perimeter of regular polygons

17 Find the perimeter of the following shapes:

a



b



18 Find the perimeter of the following regular polygons:

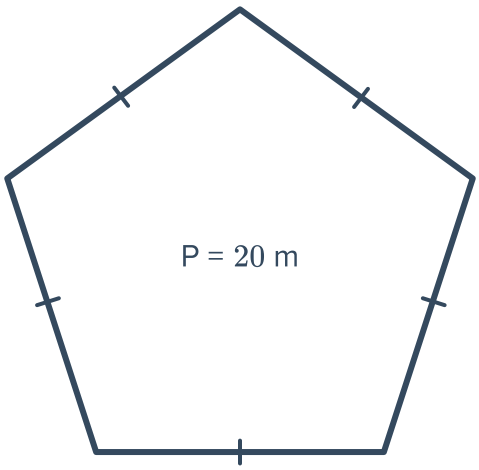
a A regular quadrilateral where the length of each side is 7 cm.

b A regular pentagon where the length of each side is 6 mm.

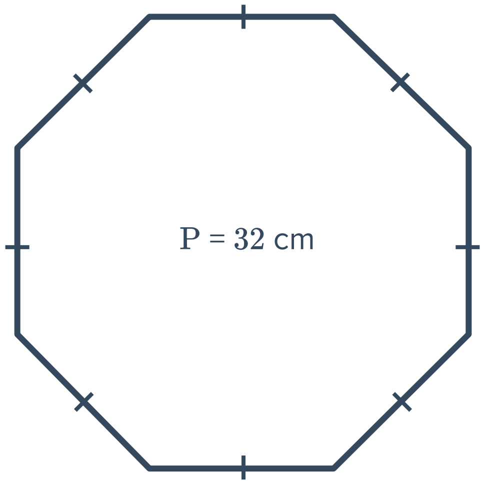
c A regular octagon where the length of each side is 6 m.

19 Find the side length of the following shap[redacted]here P is the perimeter:

a



b



20 Find the side length of the following regular polygons:

a A regular hexagon with a perimeter of 66 cm.

b An equilateral triangle with a perimeter of 21 m.

c A regular decagon with a perimeter of 150 mm.

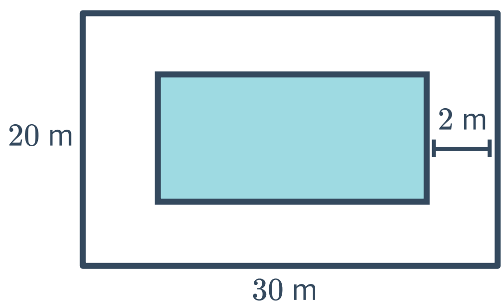
Applications

21 Hebah is tiling her bathroom with equilateral triangle tiles. She knows the perimeter of each tile, but she wants to find the length of each side. If one tile has a perimeter of 36 cm, how long is each side?

22 A swimming pool has a 2 m wide path around its edge. The outer width of the path is 20 m and the length is 30 m, as shown in the figure:

a Find the dimensions of the pool.

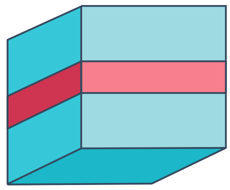
b Calculate the perimeter of the pool.

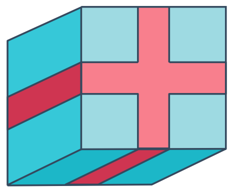


23 A square has the same perimeter as an equilateral triangle. If the triangle has sides of length 8 mm, find the side length of the square.

24 Nadia wants to build a thin wire frame for a photo that is 13 cm long and 6 cm wide. How much wire will she need to go around the entire photo?

25 A present is contained in a cube shaped box. Ribbon is wrapped around the present as shown in the first figure:





a If the side length of the box is 13 cm, find the shortest length of the ribbon needed to neatly go around the box without overlap.

b The bow requires 17 cm of ribbon. How much ribbon is needed altogether to wrap the box in this way?

c Find the total length of ribbon needed if we wrap the present with two lengths of ribbons, as shown in the second figure. Assume a single bow is tied.